ioGAS training
Fundamentals Course

UBC November 30th 2018



Overview

- The topics to be included in this session are listed below; dependent on time and participant experience and interest, some topics will be covered in more detail than others.

- Data Import
- Column Properties and Aliasing
- User Interface
- Attribute Manager
- Polygon Select tool
- Data Validation
- Attribute map
- Histograms
- Probability plots
- Box plots
- Data levelling
- Gridding
- Line plots
- 3D spatial plots
- Scatter plots
- SPLOM
- Ternary plots
- Classification diagrams
- Plot enhancements

© REFLEX 2017 ioGAS Fundamentals to Advanced

USER INTERFACE AND COMMON TOOLS OVERVIEW

DEMO DATA



Ribbon Toolbar

- Introduced in ioGAS 7.0



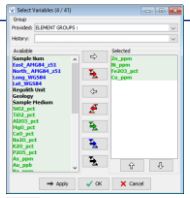
- Some bands such as Common and Window are reused on multiple ribbons:



- Quick access toolbar contains tools that are relevant in any context: Help, Attribute Legend and the list of open windows.
- Double-click on an open tab to collapse/re-instate the ribbon.
- New keyboard shortcuts (See Help File)

Selecting Variables

- Click on the icon, located on the **Tools** band of the **Home** ribbon and on the **Common Band** on most of the other ribbon toolbars.
  - In the *Select Variables* dialog, the variables are colour coded according to type.
- Use the arrows to select variables from the **Available** window and move them across to the **Selected** window.



- Select highlighted variables
- Select all variables
- Unselect highlighted variables
- Unselect all variables
- Select all aliased variables
- Select all numeric variables
- Select all text variables

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Selecting Variables Notes

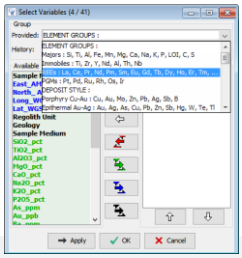
- To select multiple variables, use shift+click (for continuous selection) or ctrl+click.
- The order of variables in the **Selected** window can be modified using the **Up** and **Down** arrows or by dragging to a new location with the mouse.
  - Highlight the variable to be moved, then use the arrows to move it up or down in the order.
- The order of selected variables is very important in ioGAS – each plot will select variables from the top of the list i.e. for the variables shown above:
  - Scatter plot will be Cu (horizontal) versus Zn (vertical)
  - Ternary plot will include Cu (top), Zn (left) and Pb (right).



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Provided Variable Groups

- ioGAS provides a number of commonly used element groupings to make selecting variables easier.
- Groupings include (some examples provided below):
  - Element groups
    - Majors, immobiles, REEs, PGMs
  - Deposit style
    - Porphyry Cu-Au, epithermal Au-Ag, Ni laterite, VHMS, MVT, iron ore, sedimentary U-V-Cu
  - Secondary scavenging
    - Fe oxides, Mn oxides
  - Goldschmidt
    - Chalcophile, lithophile, siderophile

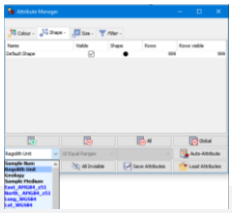


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Attribute Manager

- To open the Attribute Manager, click the icon from the Tools band on the **Home** ribbon.
  - It is also available from the **Common Band** on most of the other ribbon toolbars or using the keyboard shortcut Ctrl+A.
- Colour, shape and size attributes can be independently applied.
- To colour points by a text column, select it from the drop down options (i.e. Method) and click the *auto-attribute* icon Auto-Attribute.

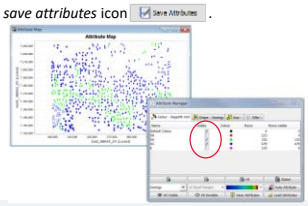
  Also do for size and shape



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Attribute Manager

- To return all points to the default black circle, click the *remove all attributes* icon All.
- Custom attribute groups or bin ranges can be created, saved and re-applied to multiple datasets.
  - To save attributes, click the *save attributes* icon Save Attributes.
- Use the checkbox next to each attribute name to select if it is visible or invisible in the plot window(s).



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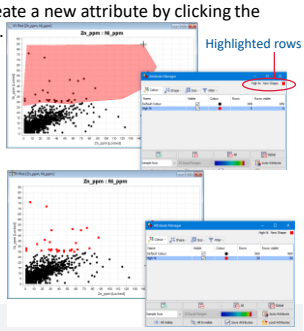
Attribute Manager Filter Tool

- Also known as the 'Super' filter.
- Use the super filter tool to filter out samples that you do not want to use in your analysis and still have 3 attributes to use during your investigation.
  - It is a way to create a subset of your data without having to open a new file.
- When a data filter is used, only data points that fit the filter criteria are displayed in any open plot window, used to calculate ranked percentiles, included in summary statistics and so on.
- The Attribute Manager Filter works in the same way as the other Attribute Manager tabs:
  - You can auto-attribute from any column in your data file
  - You can create a custom create filter and select points using the Attribute Point or Polygon tools (polygon select tool shown on next slide).

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Plot Window Tool: Polygon Select

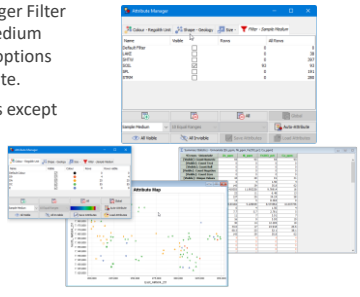
- In the *Attribute Manager*, create a new attribute by clicking the *add attribute* icon.
  - Highlight the new attribute.
  - Note: a new attribute can be created and selected on all three tabs (colour, shape, size)
- In the *Plot Window*, click the *attribute polygon* icon.
  - Create a polygon by using the left mouse button, double click to close.
  - Selected points will now be shown with new attributes.



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Attribute Manager Filter Example

- On the Attribute Manger Filter tab, select Sample Medium from the drop-down options and click Auto-Attribute.
- Make all sample types except SOIL invisible.
  - Note that now only 93 samples are visible and included in the summary stats.



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Creating a Checkpoint

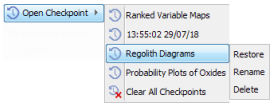
- Setup the plot windows, attribute settings and other items to save in the checkpoint.
- Select **Save Checkpoint** from the File or Home ribbon or **Save>Save Checkpoint** in all other ribbons.
  - In the ioGAS Input dialog enter a name for the checkpoint or use the default date/time stamp name assigned.



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Other Checkpoint Functions

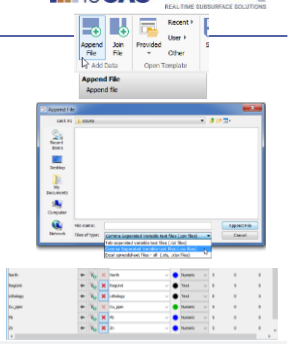
- To restore a checkpoint select from the File or Home ribbon or **Open>Open Checkpoint** in all other ribbons, go to the checkpoint you want to open and select **Restore**.
- To change the name of a checkpoint click on the **Rename** option.
- To delete a single checkpoint select it from the list and click **Delete**.
- To delete all checkpoints from an ioGAS file select **Clear All Checkpoints** from the **Open Checkpoint** list.



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Appending Data

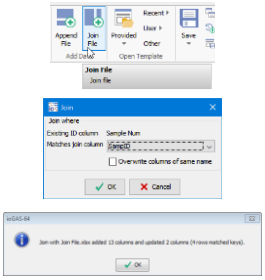
- It is possible to append data to a current ioGAS file – this is particularly useful as new data is received from the field or lab.
  - Tab or comma separated data or excel spreadsheets can be appended
  - The columns must have the same element and unit combination as the currently open dataset
    - Any columns that do not have an exact name match are denoted with a red cross.



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Joining Data

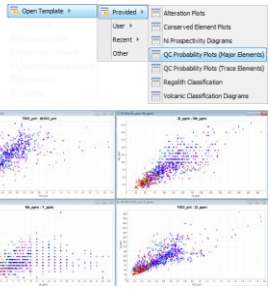
- New in ioGAS 7.0
- We can add new columns or update existing columns in the currently open dataset in ioGAS.
- Join File requires a unique matching identifier column to be present in both the original and join datasets, e.g. Sample ID.
- Columns of the same name in the dataset can be overwritten if the checkbox in the Join dialogue is checked.



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Templates

- Templates facilitate workflow application between multiple datasets
  - Templates save groups of plots and customised settings including size and location of each window
  - Templates are stored as xml files and can be shared with other users or stored on a central server
  - A number of commonly used plot combinations have been provided



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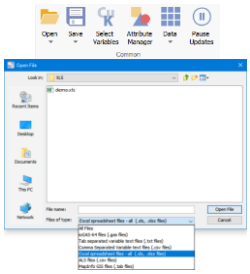
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DATA IMPORT



Import Data

- Click the **Open** button on the **File** ribbon, or any other functional ribbon (other than Help)
- Browse to the location of the training datasets.
- Use the drop-down arrow next to Files of type: and select Comma Separated Variable text files (.csv files).
  Highlight the ioGAS_Fundamentals_v6.xls file and click on the **Open File** button.



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Assigning Column Properties

- The Column Properties dialog will open and ioGAS will automatically guess the columns based on the column titles.
- Select the 'special columns' from the available drop downs and check that ioGAS has allocated them correctly.
  - Note this file contains surface sample data with no elevation values recorded.

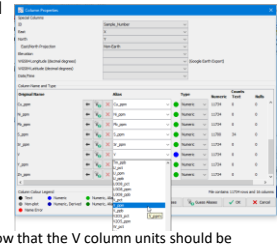


- The lower section of the dialog is where columns can be removed and column types and aliases set (see next slide).

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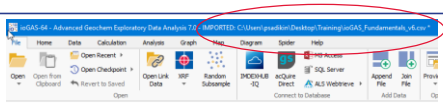
Aliasing

- Assign all the column types and aliases for this data file.
  - Check the ioGAS guesses and ensure each column is matched correctly.
  - In this dataset you want to pay attention to the Au column.
- To manually assign an element/unit alias name, select an alias from the drop down list.
  - In this dataset you will need to know that the V column units should be ppm.
- Click **OK** to assign the columns and aliases.

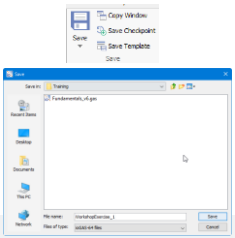


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Save File



- Note in the ioGAS title bar that the current file name is listed as IMPORTED. The file needs to be saved.
- Click on the icon on any ribbon or use the Ctrl-S keyboard shortcut.
- Type a file name in the *Save* dialog and then click **Save**.



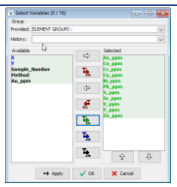
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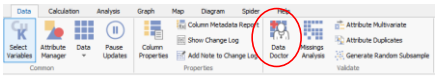
DATA VALIDATION



Data Doctor

- Use the **Select Variables** function to select all aliased elements and click **OK**.
- Select the Data Doctor from the **Data** ribbon bar.
- Work through the *Data Doctor* dialog to validate the data.

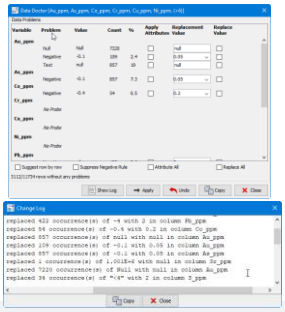




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Data Doctor Replacement Values

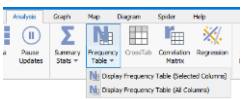
- Replace problems with ioGAS recommendations or enter your own value and tick the Replace Value box.
  - In this instance there are below detection limit, text, < signs and 'values too high' problems.
- Click the **Apply** button to apply all changes.
- The change log can be viewed and copied as a text file if desired.

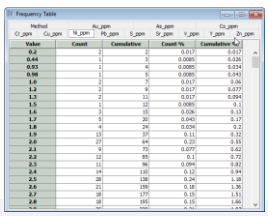


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Data Table, Summary Stats & Frequency Tables

- Select **Frequency Table** from the **Analysis** ribbon to display a frequency table for all columns. Otherwise select **Display Frequency Table (Selected Columns)** to display a table for only the variables selected.
- In the *Frequency Table* dialog, scroll through the tables for each element to look for spurious data.





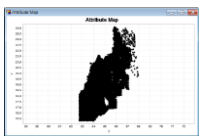
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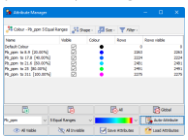
UNIVARIATE ANALYSIS



Attribute Map

- Click on **Attribute Map** from the Home or Map ribbons.
- Use the *Attribute Manager* to colour the sample points by Pb_ppm in 5 equal ranges.

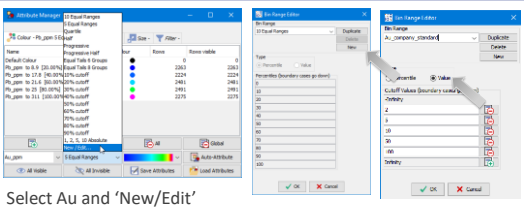




- Delete the colour attributes before you start the next slide.

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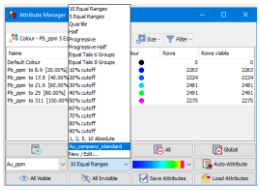
Custom attributes

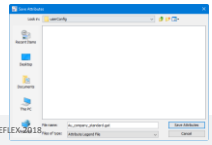


- Select Au and 'New/Edit'
- In the Bin Range Editor select 'New' and rename.
- Select 'value' and input cutoff values for Au_ppm
- Once finished hit 'OK'

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Load custom attributes

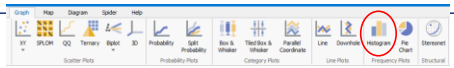




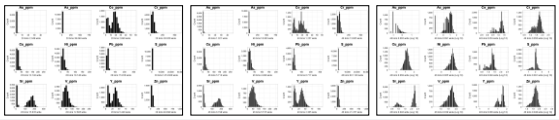
- Find New custom attribute
- Finally Auto-attribute
- Repeat for colours
- Or IF TEXT set up colours, save attribute, reload attribute, Auto-attribute, USE EXISTING COLOUR LEGEND

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Histograms

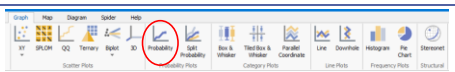


- Click on **Histogram** on the Home or Graph ribbon.
- Increase the number of bins by clicking on the *more bins* icon on the RHS toolbar.
- Apply a Log base 10 transform to the X axis by clicking the *Log X axis* icon on the RHS toolbar.



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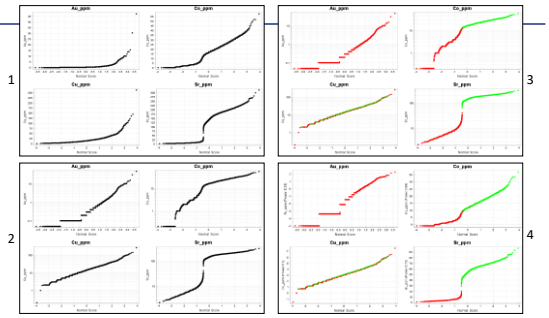
Probability Plots



- Use the *Select Variables* dialog to select Au, Co, Cu, and Sr.
- Click on **Probability Plot** on the Graph or Home Ribbon.
- Apply a Log base 10 transform to the Y axis by clicking the *Log Y axis* icon on the RHS toolbar.
- Use the *Attribute Manager* to colour the samples according to analytical method.
- Apply a power transform to the Y axis by clicking on the *De-skew Y axis* icon on the RHS toolbar.

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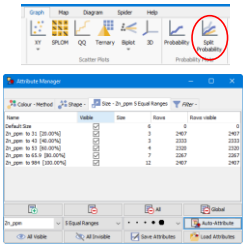
Resulting Probability Plots



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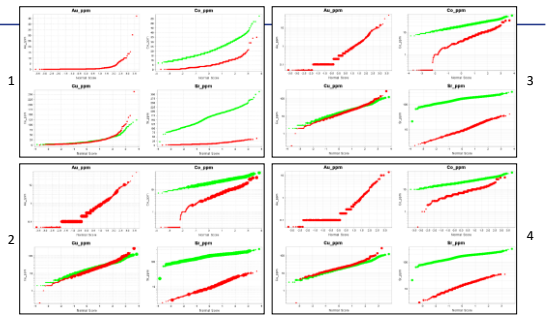
Split Probability Plots

- Click on **Split Probability Plot** on the Graph or Home Ribbon.
- Display the attribute legend by clicking the icon.
- Use the *Attribute Manager* to size the points by Zn_ppm in 5 equal bins.
- Change the point size for the highest bin to be 8.
- Remove the lower 40% of data (by Zn value) from the plot window.
- Reset all the attributes before starting the next exercise.



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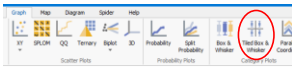
Resulting Split Probability Plots

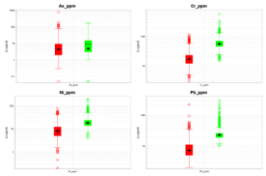


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Tukey Box and Whisker

- Select As, Cr, Ni and Pb and colour by method.
- Click on **Tiled Box & Whisker** on the Graph ribbon or **Box Plots** on the Home ribbon.
- Toggle the icon to change the zoom (zoom to outliers).
- Apply a Log base 10 transform to the Y axis by clicking the *Log Y axis* icon on the RHS toolbar.

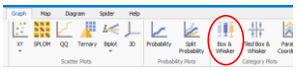




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Tukey Box and Whisker, common Y axis

- Click on **Box & Whisker** on the Graph ribbon or **Box Plots>Box Plots (Common Y Axis)** on the Home ribbon.
- Log the Y axis by clicking the *Log Y axis* icon on the RHS toolbar.
- Display the total number of samples in each category by clicking the *Show total in box* Σ icon.
- Hover over one of the 'boxes' to view the statistics for that element group.

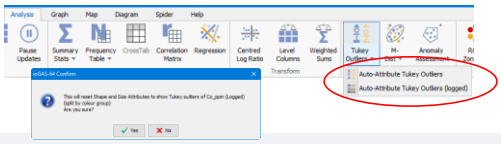




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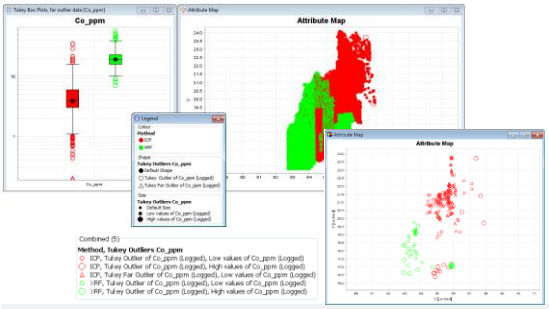
Plot Linking

- Use the *Select Variables* dialog to choose Co_ppm.
- View the Tukey box plot and log the Y axis.
- Select **Attribute Map** from the Home or Map ribbons.
- Select **Tukey Outliers > Auto-Attribute Tukey Outliers (logged)** on the Analysis ribbon. Confirm the attribute reset.
  - Note where the Co Tukey outliers are on the spatial plot.
- On the Attribute Map, display only the Tukey outliers without the zoom / scale of the map adjusting.



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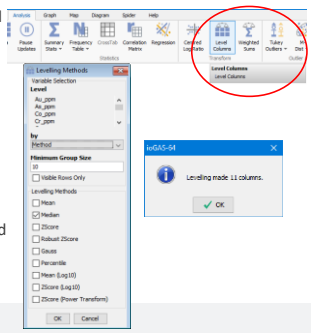
Resulting Linked Plots



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Data Levelling

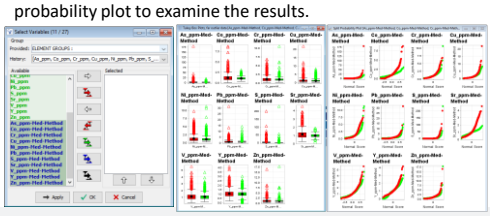
- Remove the auto-attributed sizes and shapes.
- Select all aliased elements except Au.
- Select **Level Columns** from the Transform band on the **Analysis** ribbon.
- In the *Levelling Methods* dialog, use the *by* drop down to select to level by Method.
- Choose the levelling method to be **Median** and click **OK**.



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Data Levelling Results: Median

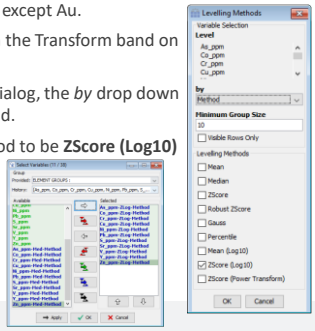
- Select the levelled data from the *Select Variables* dialog.
  - Note that ioGAS automatically names levelled data columns according to the method and the group.
- Plot the levelled data columns in a Tukey box plot and a split probability plot to examine the results.



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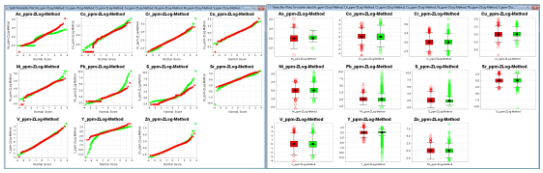
Data Levelling, Log Z-score Method

- Select all aliased elements except Au.
- Select **Level Columns** from the Transform band on the **Analysis** ribbon.
- In the *Levelling Methods* dialog, the *by* drop down to select to level by Method.
- Choose the levelling method to be **ZScore (Log10)** and click **OK**.
- Select the levelled data results and plot a Tukey box plot and a split probability plot.



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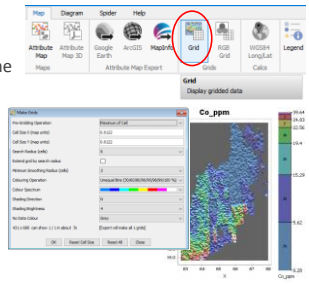
Data Levelling Results: Log Z-score



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Gridding

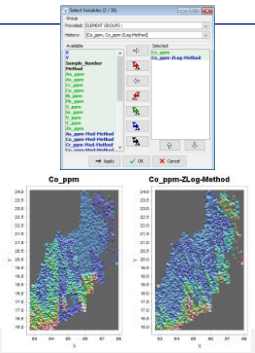
- Select Co_ppm from the *Select Variables* dialog.
- Select **Grid** from the Home or Map ribbons.
- In the *Make Grids* dialog, leave the default settings and click **OK**.
- Once the grid has plotted, open the legend by clicking the *Show/Hide Grid Legend* icon on the RHS toolbar.



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Gridding Example: Levelled Data

- Close the grid plot window.
- Select Co_ppm and Co_ppm-Zlog-Method from the *Select Variables* dialog.
- Select **Grid** from the Home or Map ribbons.
- In the *Make Grids* dialog, leave the default settings and click **OK**.
- Export to GIS



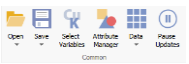
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MULTIVARIATE ANALYSIS AND PLOT ENHANCEMENTS



Introduction

- For the next section we will utilise the .xls file named Enhancements.
- Data over *Finland.*
- Click the **Open** button on the **File** ribbon, or any other functional ribbon (other than Help)
- Navigate to the Enhancements.xls file.
- Go through the 'Import data' and 'Data validation' steps



- Note Ni – do summary stats and prob plots for both, then reselect correct

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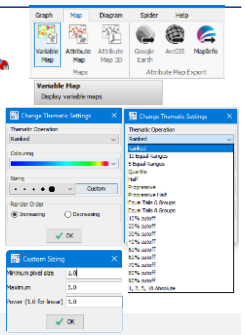
Overview

- This section will overview the map and plot types available in ioGAS to investigate and analyse data in a multivariate sense, along with commonly-used functions and enhancements such as:
  - Ranked variable maps
  - XY scatter plots
  - SPLOM
  - Ternary plots
  - Data density visualisation
  - Creating attributes, polygon groups, new variables
  - Displaying mineral and rock nodes
  - Classification diagrams
  - Custom classification diagrams
  - Calculations
  - Spider plots

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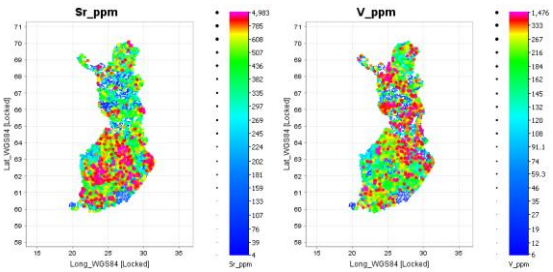
Ranked Variable Maps

- Click on **Variable Map** on Home or Map ribbon.
- Click the *Change Thematic Settings* icon on the context toolbar.
- Choose **Ranked** from a list of Thematic Operation.
- In the *Choose Colour and Size Rank Mappings* dialog, click **Custom Sizing**.
- Change the Maximum Pixel Size to be 5 and click **OK**.
- Click the *Show/Hide Colour Bar Legend* icon on the RHS toolbar.
- Choose Sr & V first



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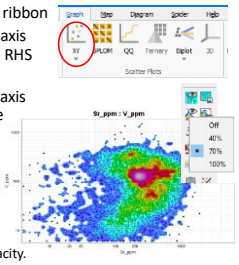
Resulting Variable Maps



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XY Plots / Scatter Plots / Cross Plots

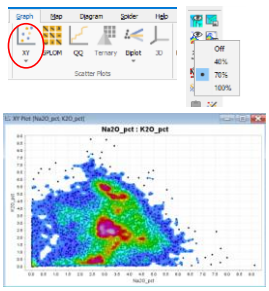
- Select Sr_ppm and V_ppm from the *Select Variables* dialog.
- Click **XY Plot** on the Graph or Home ribbon
- Apply a Log base 10 transform to the X axis by clicking the *Log X axis* icon on the RHS toolbar.
- Apply a Log base 10 transform to the Y axis by clicking the *Log Y axis* icon on the RHS toolbar.
- Display the point density by clicking the Show/Hide Point Density icon on the RHS toolbar.
  - Investigate what happens clicking this icon a number of times.
  - Right-click on the icon to change cloud opacity.



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XY Plots / Scatter Plots / Cross Plots

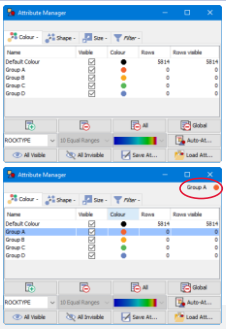
- Select Na2O_pct and K2O_pct from the *Select Variables* dialog.
- Click **XY Plot** on the Graph or Home ribbon
- Display the point density by clicking the Show/Hide Point Density icon on the RHS toolbar.
  - Investigate what happens clicking this icon a number of times.
  - Right-click on the icon to change point density display opacity.



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Creating Attributes

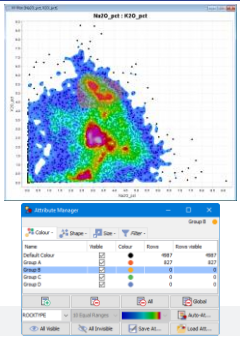
- Open the *Attribute Manager* and create 4 new colour attributes. Name them by double clicking in the Name column and typing a new name.
  - In the example, names of GroupA, GroupB, GroupC and GroupD are used.
- Highlight the GroupA row by clicking in it once (note, it is now displayed at the top of the attribute table dialog).



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Creating New Groups

- Click the *Attribute Polygon* icon on the context toolbar and digitise around the GroupA group.
  - (the polygon has been extended in this view to highlight the coloured shading).
  - Hold down Shift key to curve selection polygon.
- Open the *Attribute Manager* and see that a number of samples have been added to GroupA group.
- Deselect the GroupA group and select the GroupB group.



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Digitising Groups

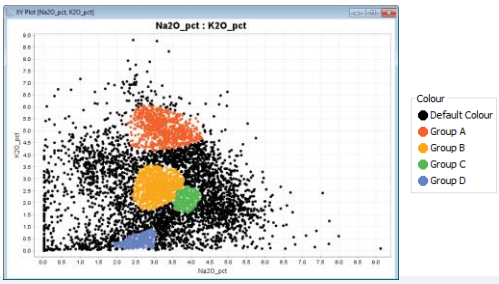
- Note that the *Attribute Polygon* icon on the RHS toolbar is still active; digitise around GroupB.
- In the *Attribute Manager*, select GroupC and digitise around these samples in the plot window.
- Finally, select GroupD and digitise those samples.
- Deactivate the *Attribute Polygon* icon on the RHS toolbar.



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New Group Results

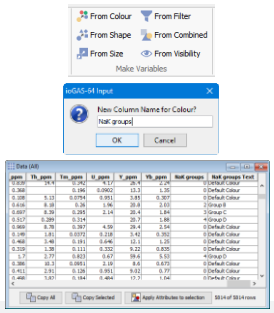
- Visualise the groups on the XY plot. Display the attribute legend by clicking the icon.



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Creating New Variable

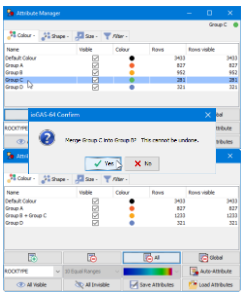
- Click Make Variable **From Colour** from the **Make Variables** band on the **Data** ribbon.
- In the *ioGAS Input* dialog, type a name for the column, such as 'NaK groups' and click **OK**.
- View the Data Table by selecting **Data** from any ribbon (except File and Help).
  - Note the colour attributes in the first column and the new variable 'NaK groups' in the last two columns (one numeric and one text variable).



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Working with Attribute Groupings

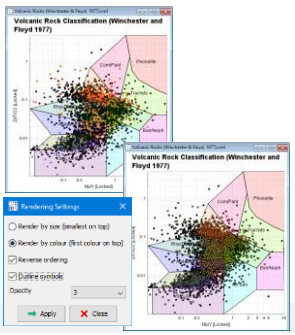
- Want to see what the attributes look like if you didn't separate GroupB and GroupC?
- In the *Attribute Manager*, click and hold on the name GroupC and drag it on top of GroupB.
- Click **Yes** to confirm the merge.
- The two groups are now merged.



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Rendering

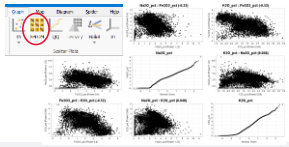
- Make the Winchester and Floyd Volcanic Rocks diagram active.
- Click the *Configure Render Settings* icon on the RHS toolbar.
- In the *Configure Render Settings* dialog, select Render by Colour and tick Reverse ordering.
- Tick on Outline symbols to display black outlines around symbols.



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Scatter Plot Matrix: SPLOM

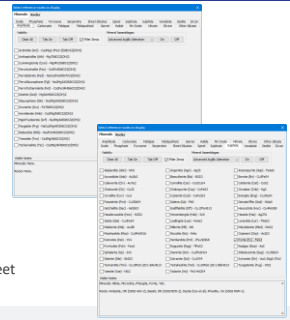
- Remove all attributes using the Global Remove icon in the *Attribute Manager*.
- In the *Select Variables* dialog, choose Fe_2O_3, Na_2O, SiO_2, K_2O, Cr.
- Click on **SPLOM** on the Home or Graph ribbons.
- Apply a power transform to both axes by clicking on the *De-skew both axes* icon on the RHS toolbar.

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Selecting Minerals and Rock Nodes

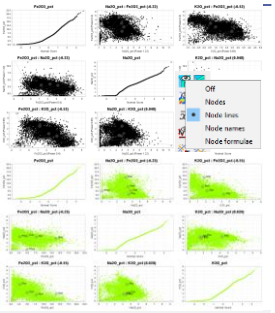
- Open the *Select reference nodes to display* dialog by clicking the *Mineral and Rock Node Settings* icon on the context toolbar.
- Click the **Clear All** button to deselect any minerals.
- Toggle on:
  - Rocks - Andesite, BCR-2 Basalt, Rhyolite (USGS Silicate tab) and Dacite (Cox et al tab)
  - Minerals - Albite and Microcline (Feldspar); Phengite and Talc (Sheet Silicates); and Pyrite (Sulphide).



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Displaying Mineral and Rock Nodes

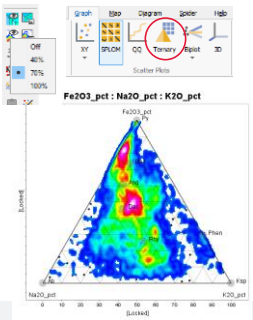
- Make the Fe-Na-K SPLOM the active plot and select the *Show/Hide Mineral and Rock Nodes* icon on the RHS toolbar.
- Click the icon again to change the mineral nodes to mineral composition lines.
  - Right-click on to choose between available mineral or rock node displays
  - You may like to remove the power transform by clicking the de-skew icon again.
  - You may like to change the default colour to be able to read the mineral labels.



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Ternary Plots

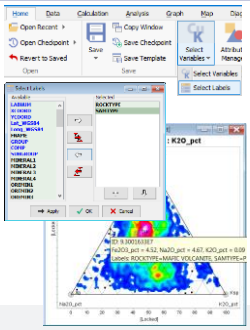
- Click on Ternary Plot on the Graph or Home ribbon.
- Turn on level 3 data density shading by right-clicking on the Show/Hide Point Density and selecting 100%.
- Display the mineral nodes by clicking the *Show/Hide Mineral and Rock Nodes* icon.
  - What happens if you click this icon again?



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Selecting Labels

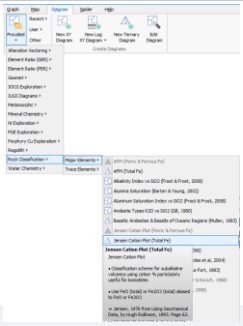
- Select **Select Labels** from the **Select Variables** drop-down menu on the Home ribbon.
- In the *Select Labels* dialog, select the properties you would like listed in the tooltip and click **OK**.
  - i.e. properties such as rock type or sample type may be appropriate.
  - When working with your own data, other properties such as prospect, project, batchID, drillhole name and so on may be useful.
- In the Fe-Na-K ternary plot, hover over a data point to see the additional labels.



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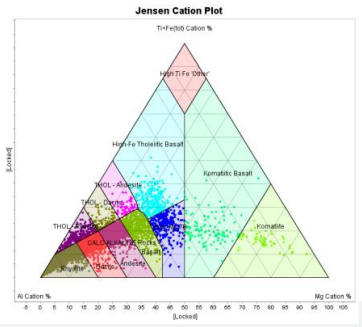
Classification Diagrams

- On the **Diagram** ribbon, click on **Provided> Rock Classification>Major Elements>Jensen Cation Plot (Total Fe)**.
  - ioGAS calculates the required fields 'on-the-fly' - if the diagram has been created in the project previously, ioGAS will ask to confirm to overwrite the columns.
- Colour the sample points according to the Jensen classification by clicking the *Colour Rows by Polygon* icon on the RHS toolbar.



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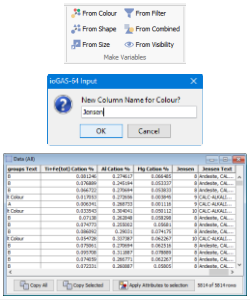
Classification Diagrams



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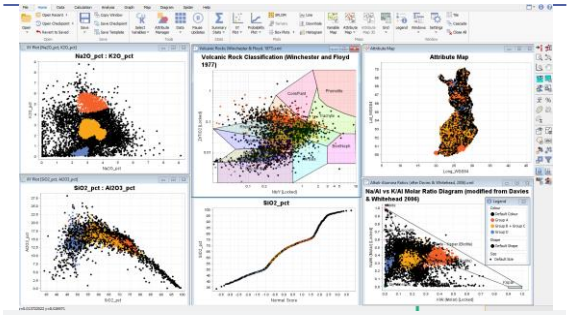
Creating New Classification Variable

- Click Make Variable **From Colour** from the **Make Variables** band on the **Data** ribbon.
- In the *ioGAS Input* dialog type a name for the column, such as 'Jensen' and click **OK**.
- View the Data Table by selecting **Data** from any ribbon (except File and Help).
  - Note the new variable 'Jensen' in the last two columns (one numeric and one text variable).



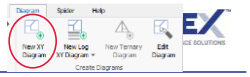
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More classification diagrams…

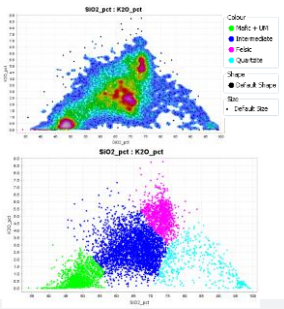


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DIY Classification Diagram



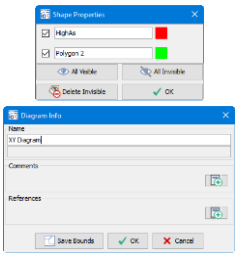
- Select variables Si & K
- Classify into groups, save new colour group
- Select label > Geology and check
- Click **New XY Diagram** from Create Diagrams band on **Diagram** ribbon.
- Use draw polygon tool to create diagram
- Hold down S while drawing to snap to point
- Change the label and colours with the Change Shape Properties icon
- Play with the icons in the RHS menu bar
- Save diagram and reopen



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Editing the Diagram

- The properties of each shape can be edited using the *Shape Properties* icon .
- Edit the name of the Polygon and the colour.
- The title of the diagram can be edited using the *Edit Diagram Properties* icon .



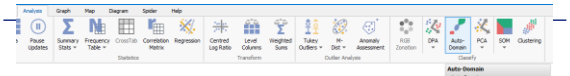
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Saving and Using Custom Diagrams

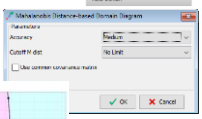
- The diagram can be saved using the *Save Diagram As* icon .
  - The default location is in your roaming app data folder, but you can select to save them anywhere.
- Diagram files can then be shared between people and ioGAS files.
- The predominant use for custom diagrams is to allocate rows to polygons, using the *Colour Rows by Polygon* icon and making a new variable from the colour attribute.
  - New data can then be imported into the ioGAS file and allocated to the custom polygon areas.

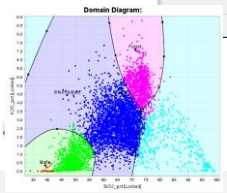
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Auto-domain classification diagram



- You can re-assign groups based on the classification diagram using the 'Colour rows by polygon' icon
- Play with the options
- *Save file for later!*

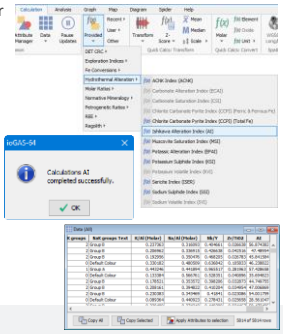




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Calculations: Provided

- On the **Calculation** ribbon, under Run Calculations band, click on **Provided>HT Alteration>Ishikawa Alteration Index**
- In the *ioGAS* dialog, click **OK** to confirm that the calculation is complete.
- Click the View Data (all columns) icon to open the data table and view the new column.
- Click the Display Ranked Variables Map icon to see the spatial distribution of AI data.

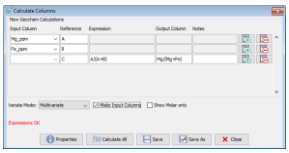


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Calculations

- Select **New Geochem Calculation** from the Calculation ribbon.
- In the *Calculate Columns* dialog, click the green plus sign icon to add another input column.
- Select Mg_ppm as the first input column. Select Fe_ppm as the second input column.
  - Note the molar units are toward the end of the drop down list or tick the *Show Molar only* option.
- Add a new input column, but leave the field empty. In the Output Column field, name the column Mg/(Mg+Fe).
- Type in the Expression field: A/(A+B).
- Tick *Make Input Columns*.
- Click **Calculate All**.

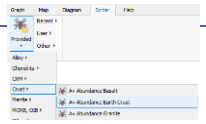
Could do Cu tenors, REE ratios, alteration indices

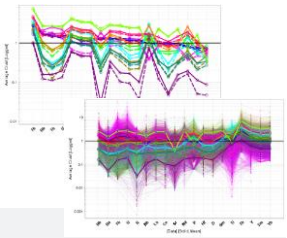


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Spider Plots

- On the Open Spider band under the **Spider** ribbon, click on **Provided>Crust>Av Abundance Earth Crust**
- Clicking the *Change Visibility of Data Points* icon will display a line for each sample point on the Spider diagram.
- Click the *Change Percentage Summary* icon on the RHS toolbar.
- Change the *Column Display Mode* by clicking on the icon on the RHS toolbar.

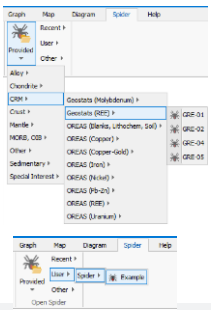




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Spider Plots: CRMs, Custom Plots

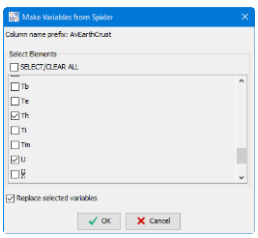
- There are also spider plots provided for QAQC purposes.
  - A number of spider plots are provided for common reference materials from Geostats and OREAS
    - These include the standards supplied with the REFLEX pXRF rental units
- It is possible to create and save your own spider plots
  - The plots are stored in XML format
  - An example of the format and syntax is provided



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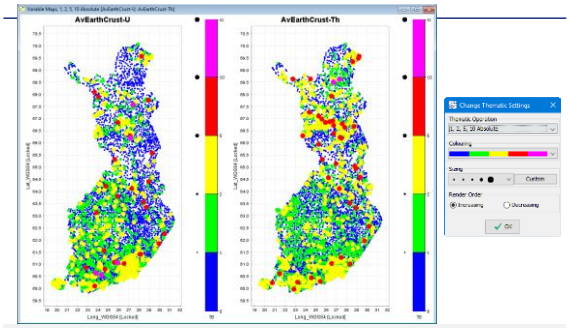
Spider Plots: Make Variables

- New in ioGAS 7.0 is the ability to add the normalized values for each spiderplot element to the dataset as new columns.
- Click on the **Make Variables from spider** icon on the context toolbar.
- Use in conjunction with variable maps and new times-background thematic operation to colour the data.



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Times-background Thematic Maps



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WORKING WITH DRILLHOLE DATA



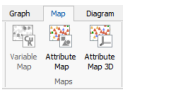
Introduction

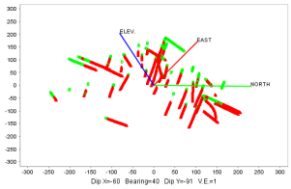
- For this section of the exercises, we will need a dataset with 3D drillhole data.
- Close the current ioGAS project (with or without saving).
- Open the 3D_Drillholes_v6.gas file.
- Go through the data import and data validation steps.
  - View the Data Table by selecting **Data** from any ribbon (except File and Help).
  - Have a look at the data available in this file.

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3D Spatial Plots

- Select **Attribute Map 3D** from the Home or Map ribbons.
- Use the Attribute Manager to also colour the samples by Ox to make them easier to see.
- Toggle the icons to look down, north and east.



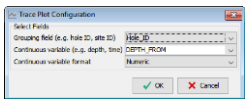


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Line Plots



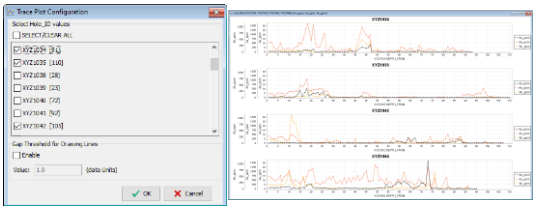
- Select Au, As and Sb in the *Select Variables* dialog.
- Click on **Line** on the Graph or Home ribbon.
- In the *Line Plot Fields* dialog, select **Hole_ID** as the *Grouping Field*, **Depth_From** as the *Continuous Variable* and **Numeric** as the *Continuous Variable Format* and click **OK**.



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Line Plot Configuration

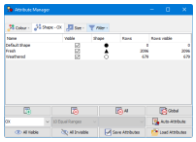
- In the *Line Plot Configuration* dialog, select drillholes XYZ 1034, 1035, 1042 and 1064 and click **OK**.

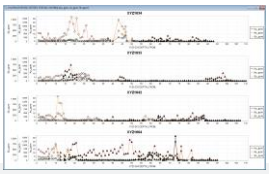


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Line Plot Manipulation

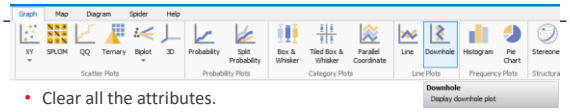
- Use the *Attribute Manager* to apply a shape attribute to the Ox column.
- Change the icon for fresh samples to be a solid triangle and weathered samples to be an open circle.
- Toggle the icons in the RHS toolbar to investigate their effect on the plot.
  - Which icon displays the triangles?



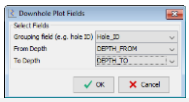


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Downhole Plots



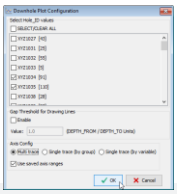
- Clear all the attributes.
- Select As, Sb, Fe and Mn in the *Select Variables* dialog.
- Select **Downhole** from the Graph or Home ribbon.
- In the *Downhole Plot Fields* dialog, select **Hole_ID** as the *Grouping Field*, **Depth_From** as the *From Depth* and **Depth_to** as the *To Depth* and click **OK**.

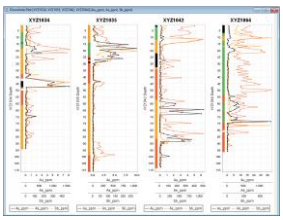


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Downhole Plot Configuration

- Select the same drillholes as the previous exercise, namely XYZ1034, XYZ1035, XYZ1042 and XYZ1064.
- Leave the default as *Multi trace* and click **OK**.





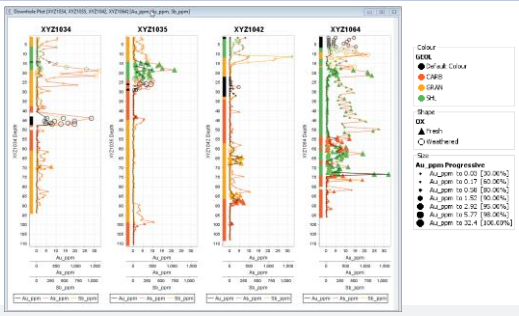
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Downhole Plot Manipulation

- Use the context toolbar icons to scale the depth axis such that all drillholes extend from 0 to 110 and the x axes so that each drillhole has the same range for each element.
- Use the *Attribute Manager* to apply a shape attribute to the Ox column.
  - Change the icon for fresh samples to be a solid triangle and weathered samples to be an open circle.
  - Toggle the *Point Attributing On/Off* icon on the RHS toolbar to display the shapes on the downhole plot.
- Use the *Attribute Manager* to apply a colour attribute to the Geol column and size by Au using the *Progressive Half* scale.
  - Resize each of the groups to your liking.
- Display the legend.

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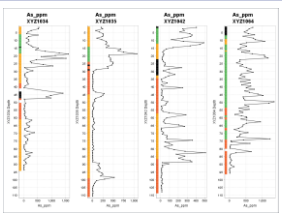
Downhole Plot Attributes



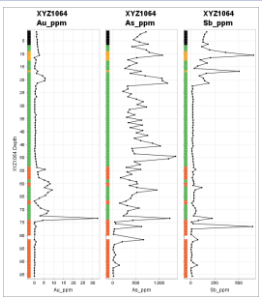
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Downhole Plot Configuration

- Repeat for *Single trace* (by group or by variable) and click **OK**.



Single-trace by Variable

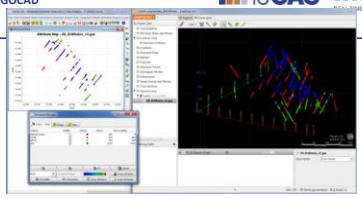


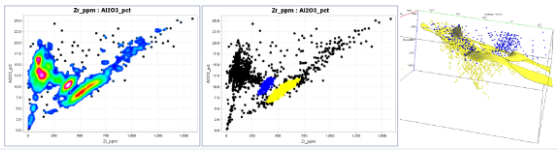
Single-trace by Group

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Another Option ... If you use Leapfrog Geo or GOCAD

LF example with YAM data





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Introduction to ioGAS Stereonet

Updated for version 7.0

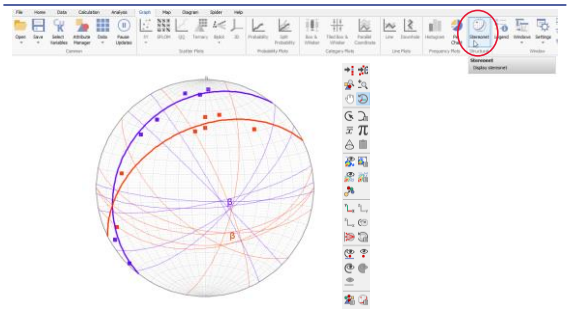


Overview of Functionality

- Planes and lines can be plotted on the same stereonet
- On-the-fly projection changes
  - Orthographic, Equal Angle (Wulff net) and Equal Area (Schmidt net)
- Place your own planes
  - User planes persist in a gas file and can be copy and pasted between files
- Calculate and display means
  - Of all data and by colour group
- Calculate pi diagram: the best fit great circle (using Bingham statistics) and β axis
- Calculate conical best fit: "small circles"
- Steep fault compensation
- Copyable statistics report including all calculations by colour group
- Point Density grid can be displayed on Stereonet plot.
- Gridded point density can be Contoured.
  - Fill point density contour with colour, forming contour regions.
  - Data points within a contour region can be attributed to colour (works similarly to Attribute Rows by Polygons in classification diagrams).

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User Interface



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RHS Toolbar Functions

	Attribute Point – Apply current attributes to selected point/s.
	Attribute Polygon – Apply current attributes to points within drawn polygon.
	Apply Attributes to All in View (first plot) - Apply current attributes to all points within the current window view.
	Zoom Dynamically - Click in the plot window and drag the cursor up to zoom out or down to zoom in.
	Pan - Click and drag using the left mouse button to pan across the data view. The middle mouse/scroll button can also be used to pan within a plot window.
	Rotate around the centre of the stereonet, in the plane of the stereonet, for Equal Angle and Equal Area plot projections. For orthographic projection, 3D rotation of the stereonet sphere (or half sphere) is around the centre of the sphere. Data points are decimated on rotation; to display all data points when rotating hold down the Alt key.

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RHS Toolbar Functions

	Place Plane (hold CTRL to place pole) – add new plane by plane line or hold CTRL key to place new plane by its pole location. Click on icon again to exit.
	Edit user planes – edit new planes generated using the Place Plane tool and/or copy and paste a number of planes at one time.
$\bar{x}$	Show Means - Display location of mean for entire dataset and visible colour groups using solid cross symbol. Cycles Mean from All rows + Groups, by Groups, or None (no Mean displayed).
π	Best fit great circle - Display best fit great circle (Bingham statistics) and beta axis for entire dataset or colour group. Cycle through options or right mouse-click and choose mode.
	Best fit small circle - Display best fit small circle (conical best fit) for entire dataset or colour group. Cycle through options or right mouse-click and choose mode.
	Show copyable report of visible stats - Copyable report window containing the results of the calculations for all currently visible statistics. These can be copied onto the clipboard and pasted into another application.
	Show/Hide Point Density: Show grid of the density of the poles to planes and lineation data. Cycle through different opacity level or right mouse-click and choose mode.
	Point Density Settings: Configuration settings for the displayed point density.

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RHS Toolbar Functions

	Show/Hide Contours: Display contour polygons, which are drawn along boundaries between grid bins. Toggle display of contours on/off.
	Contour Settings: Configuration settings for the displayed density contour.
	Attribute Rows by Bin: When contour fills are enabled, this will attribute all data points within each contour polygons.
	Reset View to XY Plane - Resets the rotation, so that the Z axis is toward the user and the N pole of the stereonet is at the top
	Reset View to YZ Plane - Resets the rotation, so that the X axis is toward the user and the N pole of the stereonet is at the top of the plot. This function is only valid in orthographic projection.
	Reset View to XZ Plane- Resets the rotation, so that the Y axis is toward the userand the N pole of the stereonet is at the top of the plot. This function is only valid in orthographic projection.

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RHS Toolbar Functions

	Show full sphere – Display the other half of the projection sphere in an orthographic plot.
	Set Rotation – Set specific amount of rotation (XYZ for orthographic, Z only for equal angle and equal area) for the plot display.
	Cycle between show all data, planar data only, linear data only or no data displayed.
	Toggle display of poles on/off.
	Toggle display of planes on/off.
	Shade Planes; Only available in Orthographic Projection mode.
	Toggle display of lineation vectors from the origin on/off.
	Change projection between Orthographic, Equal Angle to Equal Area.
	Stereonet Settings.
	Rendering settings - Rendering order options only available when **Use Attribute Manager** symbols and size options are selected in Stereonet Settings.

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The Dataset

- Data requirements
  - A column containing Dip values (valid range 0-90)
  - A column containing Dip Direction values (valid range 0-360)
    - There is support for strike measurements to be auto-converted to Dip Direction using the Right Hand Rule (RHR)*
  - A column specifying the type of structure – this is optional
    - This column is used to specify which structures are planes and which are lines
    - If this column is not utilised, all data are assumed to be planes.
- No other data formats, conventions or conversions are supported.
- * Right Hand Rule (RHR): if data are specified as strike and dip, it will be assumed that the dip is 90° to the right of the strike.

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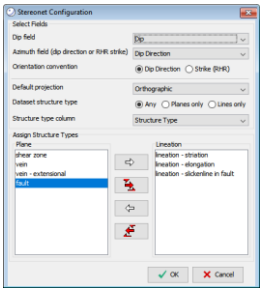
Creating a Stereonet



- The stereonet tool can be accessed by clicking on the icon from the Graph ribbon bar.
- This will open the Stereonet Configuration dialog.

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Configuring a Stereonet

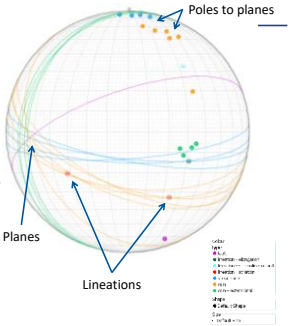


- Columns containing Dip and Dip Direction information are to be set in the appropriate fields in the Stereonet Configuration window.
  - These will be 'remembered' in the ioGAS file.
- The default projection can be specified for each stereonet.
- A structure type column is optional
  - If it is specified, structure types can then be assigned as planes or lineations.

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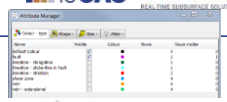
Default Visualisation

- ioGAS utilises 'special' stereonet symbols:
  - Poles to planes
  - Lineations
- Planes are displayed as lines and poles to planes.
- Lineations are shown as points.
- Colours are assigned by the attribute manager.
- The RHS toolbar is utilised to modify the visualisation parameters and perform calculations on the data.

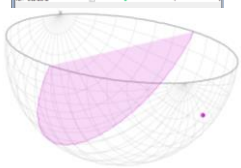


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Visualising a Single Plane



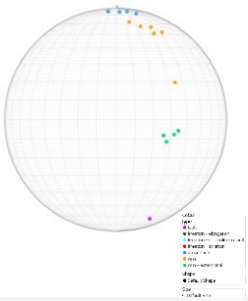
- This view is achieved by having only one row visible in the attribute manager.
- The shading of the plane is achieved by selecting the Shade planes: Off | On icon on the RHS toolbar.
- Plane shading is only activated in Orthographic Projection.
- A maximum of 15 shaded planes can be displayed on the stereonet plot unless the maximum number is changed in the Stereonet Settings dialog.



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Visualising Planes and Poles

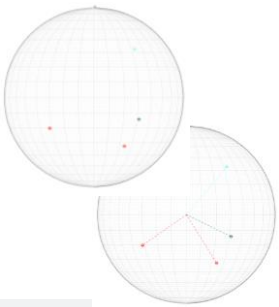
- The Show icon toggles between displaying All data, Planar data only, linear data only and no data.
- The Show Poles icon can then be used to turn poles on and off.
- The Show Planes icon can then be used to turn poles on and off.
- This image shows only the Planes displayed, poles displayed, planes off.



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Visualising Lineations

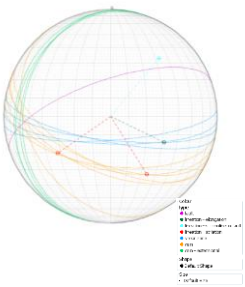
- To visualise only lineations, this can be done by clicking the Show icon twice.
- Lineation vectors from the origin can be visualised using the Lineation Vectors icon
  - Note, that lineation vectors are only displayed in Orthographic projection (the icon is greyed out in other projection modes).



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Visualising Planes and Lineations

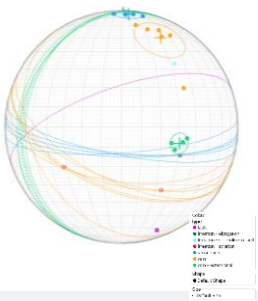
- Combinations of planes and lineations with vectors can be visualised using the various visibility icons.
  - Note, that lineation vectors are only displayed in Orthographic projection (the icon is greyed out in other projection modes).



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Determining Averages / Means

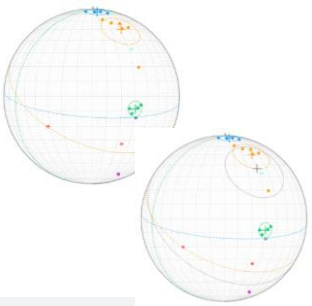
- Use the Show Means icon to toggle the means for the poles to the planes, displayed as crosses.
- A mean will be calculated for each colour group assigned in the attribute manager.
- A mean plane will be displayed with a dashed line.
- The 95% confidence circle is shown as a circle around the mean pole value.



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Visualising Averages / Means

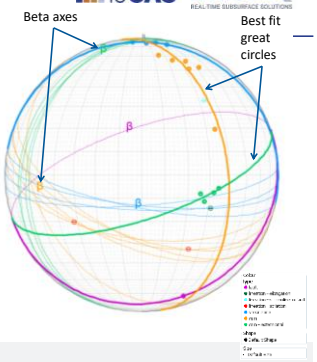
- The planes can be turned off using the Show Planes icon
- Toggling the Show Means icon will display the mean for the groups and all the data, where all data is shown in grey.



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Calculating Pi Diagrams

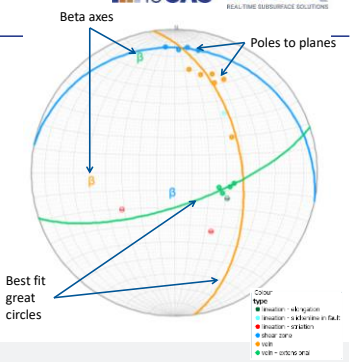
- Use the Pi Diagram icon π to calculate a best fit great circle and display the β axis.
- The best fit great circle will be displayed as a thick line.
- The beta axis (pole to the best fit great circle) is displayed as a β symbol.



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Visualising Great Circles and β Axes

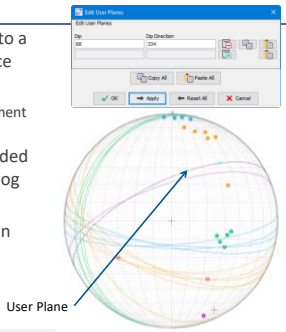
- Note that a great circle and beta axis will be calculated for each colour group, so it may be necessary to make some colour groups invisible in the attribute manager.
- It is also possible to turn off the planes when visualising the great circles.
- It is also possible to change projection on-the-fly.



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Creating and Editing User Planes

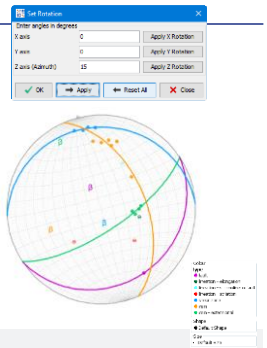
- User planes can be added to a plot window using the Place Plane icon
  - This allows interactive placement in the plot window.
- User planes can also be added in the Edit User Planes dialog using the icon
- User planes are displayed in grey.



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Set Rotation

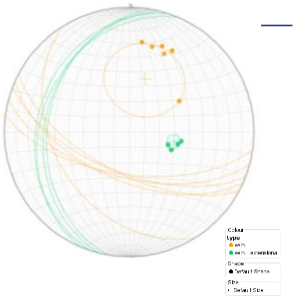
- The stereonet can be displayed with a specified rotation.
- Set a rotation using the Set Rotation icon
- For Orthographic projection, rotation around all 3 axes can be set.
- For Equal Area and Equal Angle projections, a rotation around the Z axis (Azimuth) is appropriate.

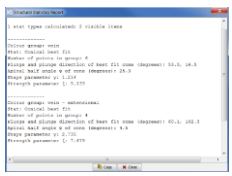


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Conical Best Fit

- To calculate the conical best fit for poles to planes, click the Conical Best Fit icon
- A conical best fit will be determined for each colour group and displayed as a dotted cross and cone outline.

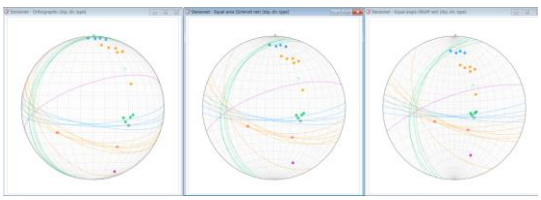




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Different Projections

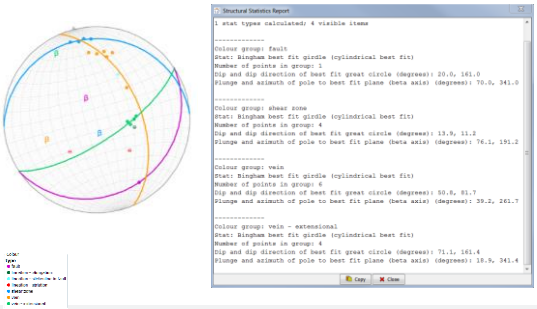
Toggle between different projections using the icon.



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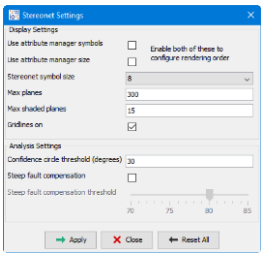
Structural Statistics Report

A copyable report can be generated for the calculated statistics currently displayed on each plot.



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Stereonet Settings

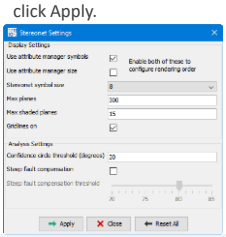


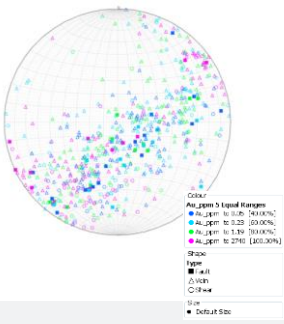
- The 'special' stereonet symbols can be overridden to use the attribute manager symbols.
- Other features that can be overridden are: maximum number of planes to display and shaded planes to display.
- Steep fault compensation can be enabled.

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Attribute Manager Override - Symbols

- To use attribute manager shapes, tick the box in the Stereonet Settings dialog and click Apply.

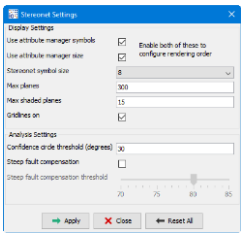


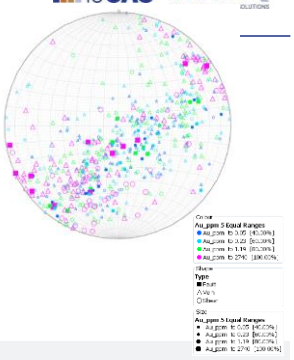


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Attribute Manager Override - Size

- To use attribute manager sizes, tick the box in the Stereonet Settings dialog and click Apply.

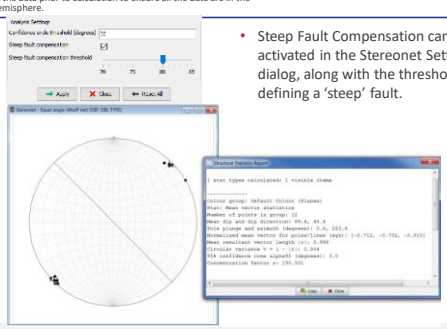




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Steep Fault Compensation

With the situation of steep faults, the mean calculation is modified by rotating the data prior to calculation to ensure all the data are in the same hemisphere.



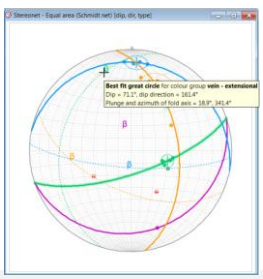
- Steep Fault Compensation can be activated in the Stereonet Settings dialog, along with the threshold defining a 'steep' fault.

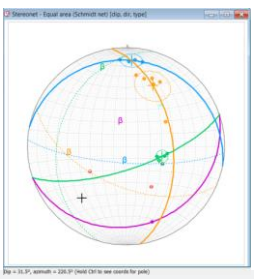
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Other Information

Hover over a pole, it will highlight the plane. Hover over calculated values (β axes, means, conical best fit) to view calculation results.

Move the cursor anywhere around the stereonet and the coordinates will be shown in the bottom left corner of the screen.

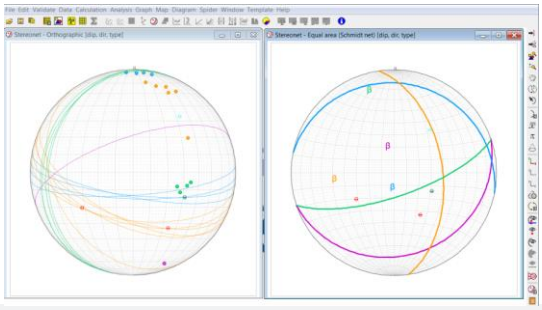




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Visualising Structural Data

It is possible to display multiple stereonets with different visualisation parameters, they work the same as any other plot window in ioGAS.

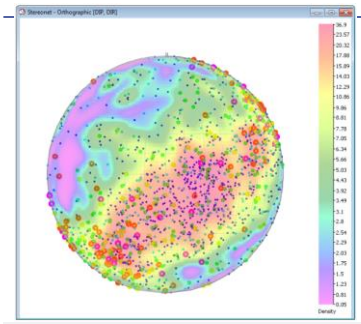


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Other Neat Tricks



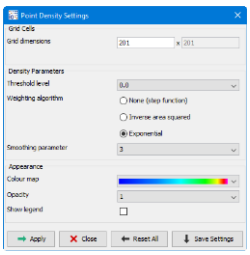
Point Density Grid



- Activated from the icon on the Right Hand Side Toolbar.
- Can be displayed with calculated statistics such as mean, median and best fit great circle.
- Point density parameters can be adjusted in the Settings menu .

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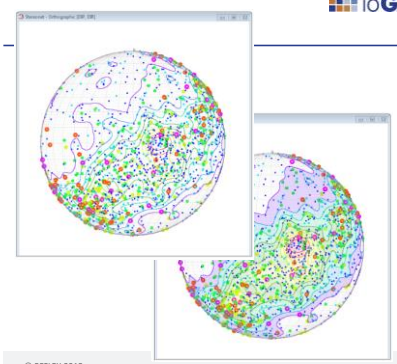
Point Density Grid Settings



- Grid Cells: This controls the size of the grid used to draw the point density map. Increase this to increase the resolution of the point density grid.
- The Density Parameters panel allows some parameters of the density algorithm to be configured.
  - Treshold level: Specifies the minimum value of the number of data points in a cell (after smoothing has been performed) which will be visible in the resulting plot.
  - Weighting algorithm: Choose between inverse area squared or exponential weighting algorithm, or no weighting algorithm applied (step function)
  - Smoothing parameter: This is a parameter to the weighting algorithm that determines the smoothness of the point density grid. 3 sigma is the default value.
- The Appearance menu controls the colour scheme of the point density grid and its opacity. Point Density Legend can be toggled on/off.
- Click on Apply to apply selected settings to the stereonet plot. Close to exit the dialogue. Reset All to change all settings back to default.

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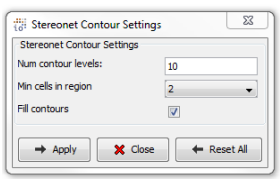
Contour Lines



- The plot on the left shows contour lines being drawn around point density bins. Toggled using the icon.
- The other plot fills the contour polygons with colour (as set in the contour settings dialogue) This allows the data points to be auto-attributed to the contour polygons.

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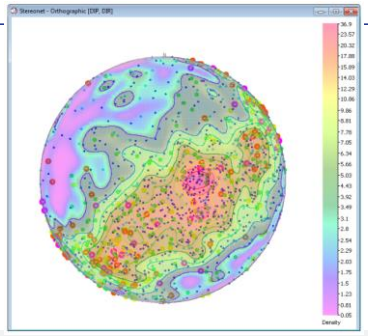
Stereonet Contour Settings



- Num contour levels: Control the maximum number of contour levels that will be drawn on the stereonet plot. Note that the actual displayed contours will be one less than the number of contour levels entered.
- Min cells in region: Define the minimum number of cells that will be used to form a contour region.
- Fill contours: Fill contour regions with colour.

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Displaying both point density grid and contour regions



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Thank you for your time

For more information contact

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